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# Trio Genetic Analysis – Exome (Germline) HG-TRIO-EXO

## Applications

Method utilizes exome sequencing of a parent–parent–child trio to identify inherited and de novo germline variants underlying genetic disorders. Protein-coding regions are captured and sequenced, followed by alignment to a reference genome, variant calling, and trio-aware analysis to assess inheritance patterns. This approach improves diagnostic yield by enabling precise filtering of candidate variants based on Mendelian segregation, making it especially effective for rare and undiagnosed conditions. Trio Exome suitable for investigation of rare cariant association, adaptive TDT models and infering transmission of inherited alleles.

Pros

- Higher diagnostic yield than singleton exome
- Accurate detection of idividual variants
- Efficient filtering of benign variants
- Ideal for rare and undiagnosed diseases
- Strong support for clinical interpretation

Notes

- Requires samples from both biological parents + individuals
- Limited to coding and captured regions
- Structural variant detection is limited to indels
- Clinical validation recommended for key findings

## Input

Genomic DNA (Germline)

DNA Quantity: 100 ng–1 µg

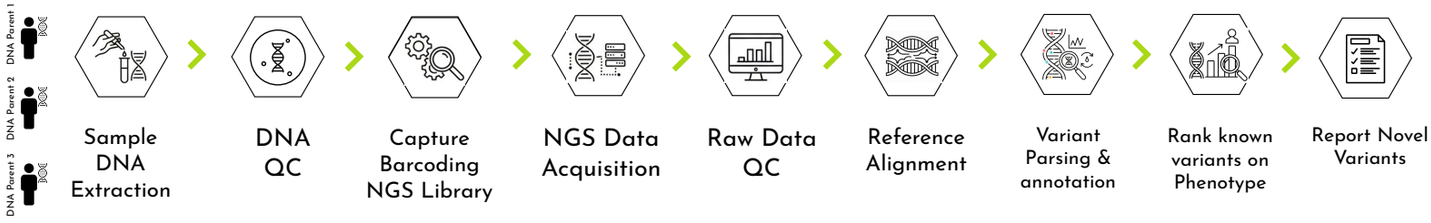
## Sample Types

Blood | Tissue | Buccal Swabs

## Platform

Illumina platforms | 150 * PE | 100 X Exome | ~35 MB x 3 x 24 GB

## Process Map




## Standard Deliverables

Download Sample Report

- Sample and Library Quality Control (QC)
- Summary of Methods and Raw Data Quality Control
- Reference Genome Alignment Summary- trio
- Variant Summary Trio > Parent 1; Parent 2 > Individual
- Variant annotation and Variant effect prediction
- Novel Variants / Known Variants / Pathogenic Variants (if identified)
- Inheritance pattern analysis - Recessive/ Dominant disorders
- Phenotype based variant ranking (if scoped)